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- Power Electronics
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- Telecom

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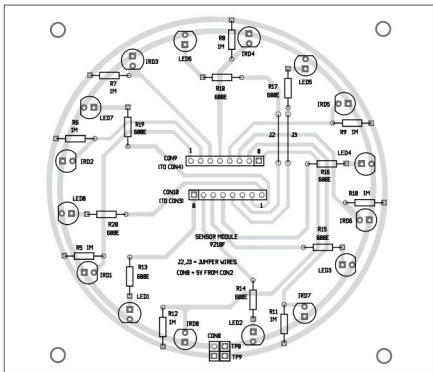


Fig. 11: Component layout of the PCB of the sensor module

PARTS LIST

Semiconductors:

- IC1, IC5 - 7805, 5V regulator
- IC2 - ATmega16 microcontroller
- IC3, IC4 - L293D motor driver
- IC6 - ATmega8 microcontroller
- IRD1-IRD8 - 5mm photodiode (IR receiver)
- LED1-LED8 - 5mm white LED
- LED9, LED10 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R13-R20 - 680-ohm
- R2, R4 - 10-kilo-ohm
- R5-R12 - 1-mega-ohm
- RNW1 - 10-kilo-ohm

Capacitors:

- C1, C6 - 10 μ F, 25V electrolytic
- C2, C3, C7, C8 - 0.1 μ F ceramic disk
- C4, C5, C9, C10 - 22pF ceramic disk

Miscellaneous:

- TX1 - 433MHz transmitter module
- RX1 - 433MHz receiver module
- Xtal - 8MHz crystal
- ANT.1, ANT.2 - 17cm single wire antenna
- S1, S3 - On/off switch
- S2, S4 - Tactile switch
- DIP1 - 4-way DIP switch
- CON1, CON5 - 2-pin terminal connector
- CON2, CON8 - 2-pin connector
- CON3, CON4, CON9, CON10 - 8-pin connector
- CON6, CON7 - 3-pin connector
- M1, M2 - 12V DC geared motor
- M3 - 12V water pump motor

Limitations

IR photodiodes also react to sunrays, just like fire, since these too contain IR waves, thus allowing the robot to make false judgments. However, the

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

effect of sunlight and other factors is not significant if we use better sensors (which are far more costly).

Future enhancements

To improve the system:

1. Use more sensors such as smoke detectors, temperature sensors and thermal image processing for feedback and to improve the accuracy and reliability of the system further.

2. Replace IR photo sensors with more dependable ones, which have filters, because IR sensors even take sunlight as a source of IR waves, leading to errors and false alarms. ●



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